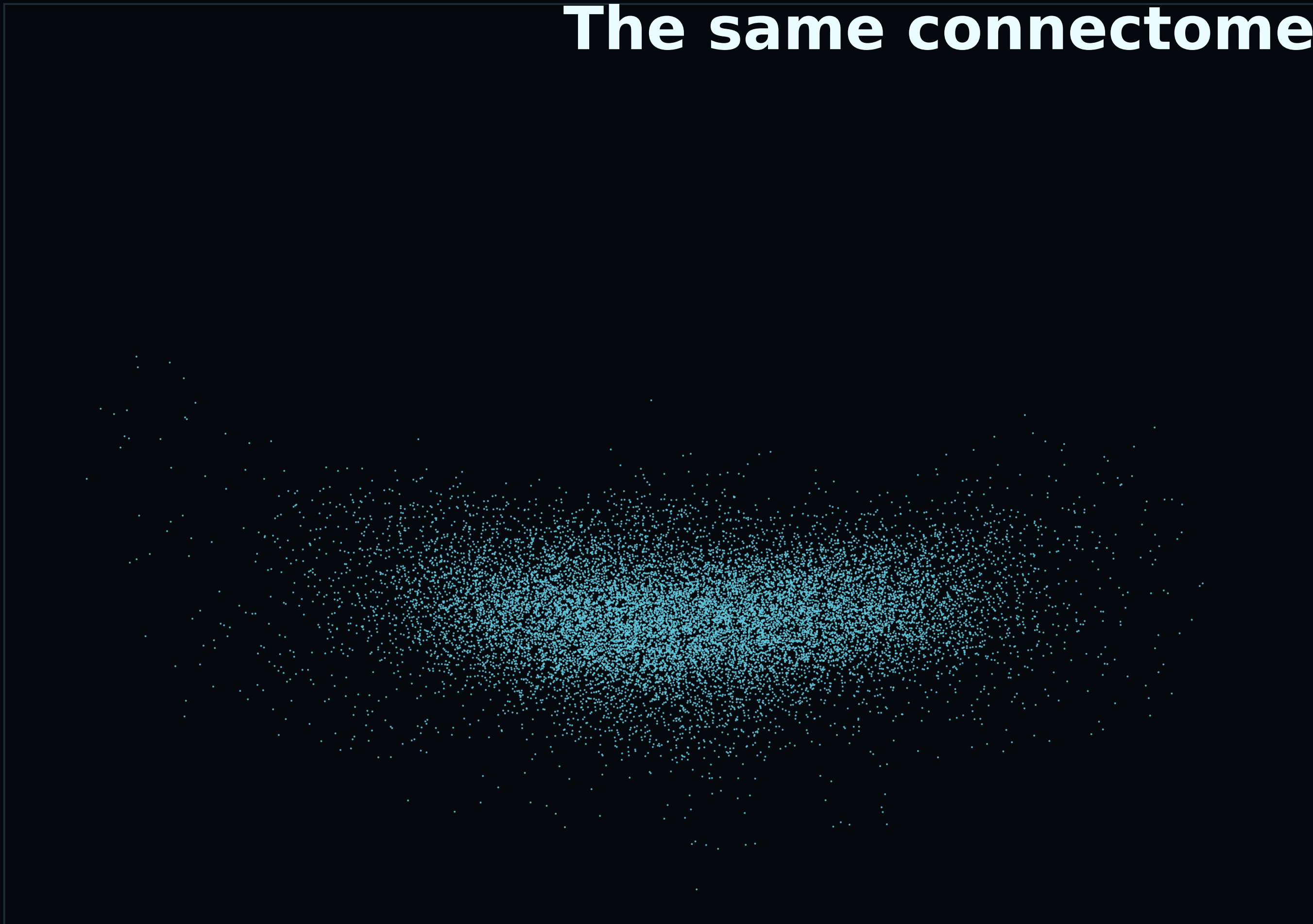
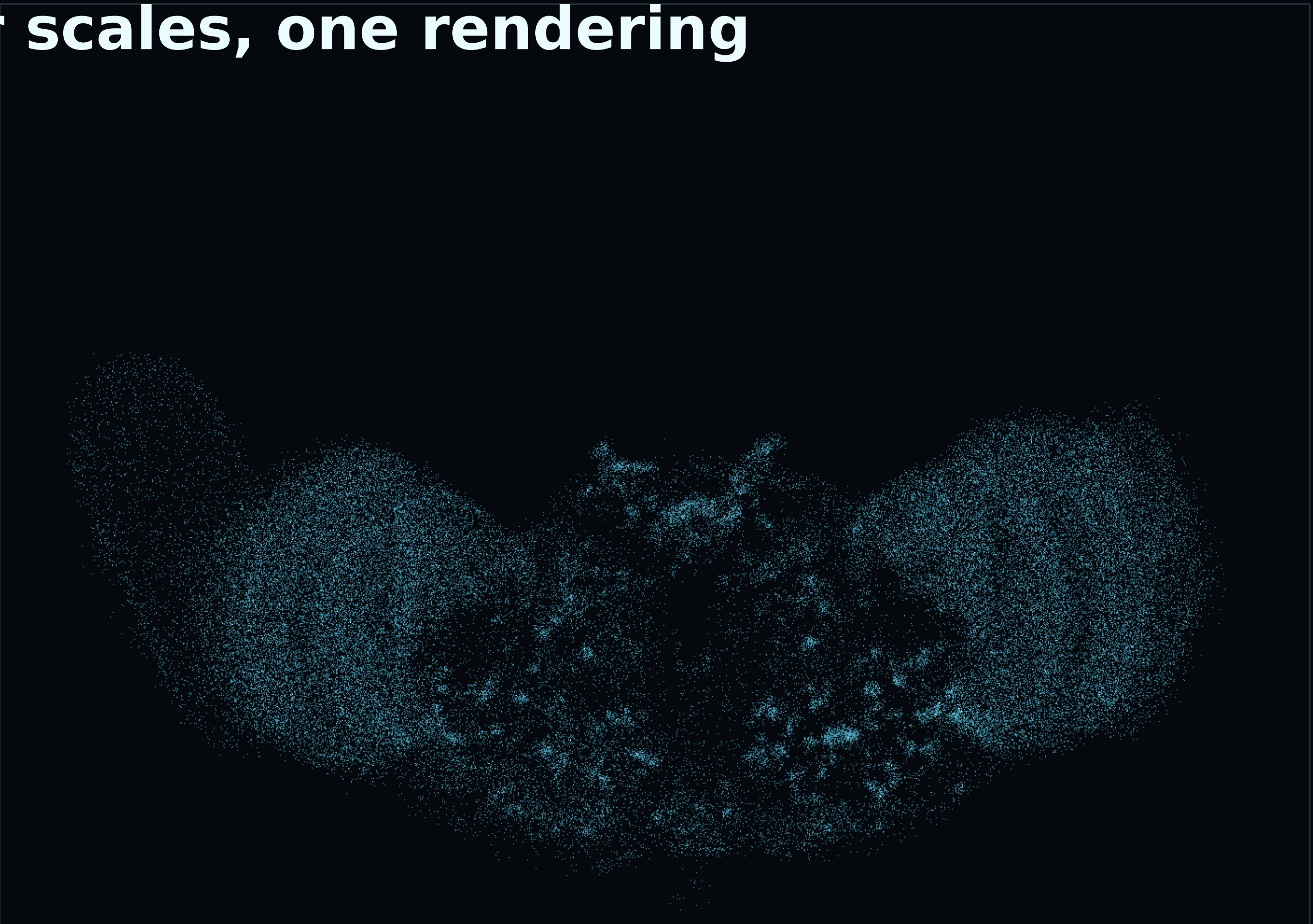


The same connectome at four scales, one rendering

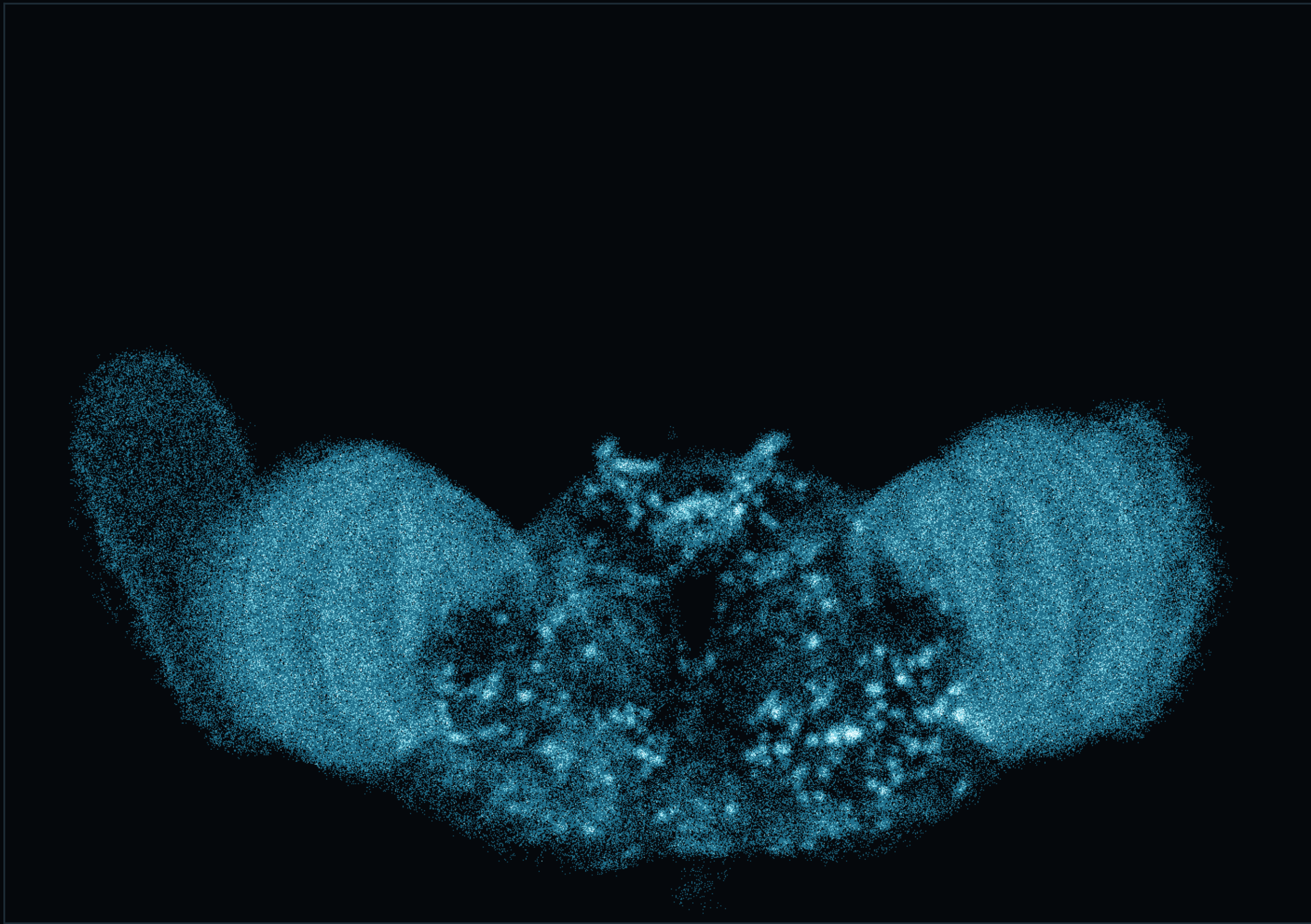


phase-5 coarse grouping: 15,997 somata coarser than one 28 nm pixel, drawn as points

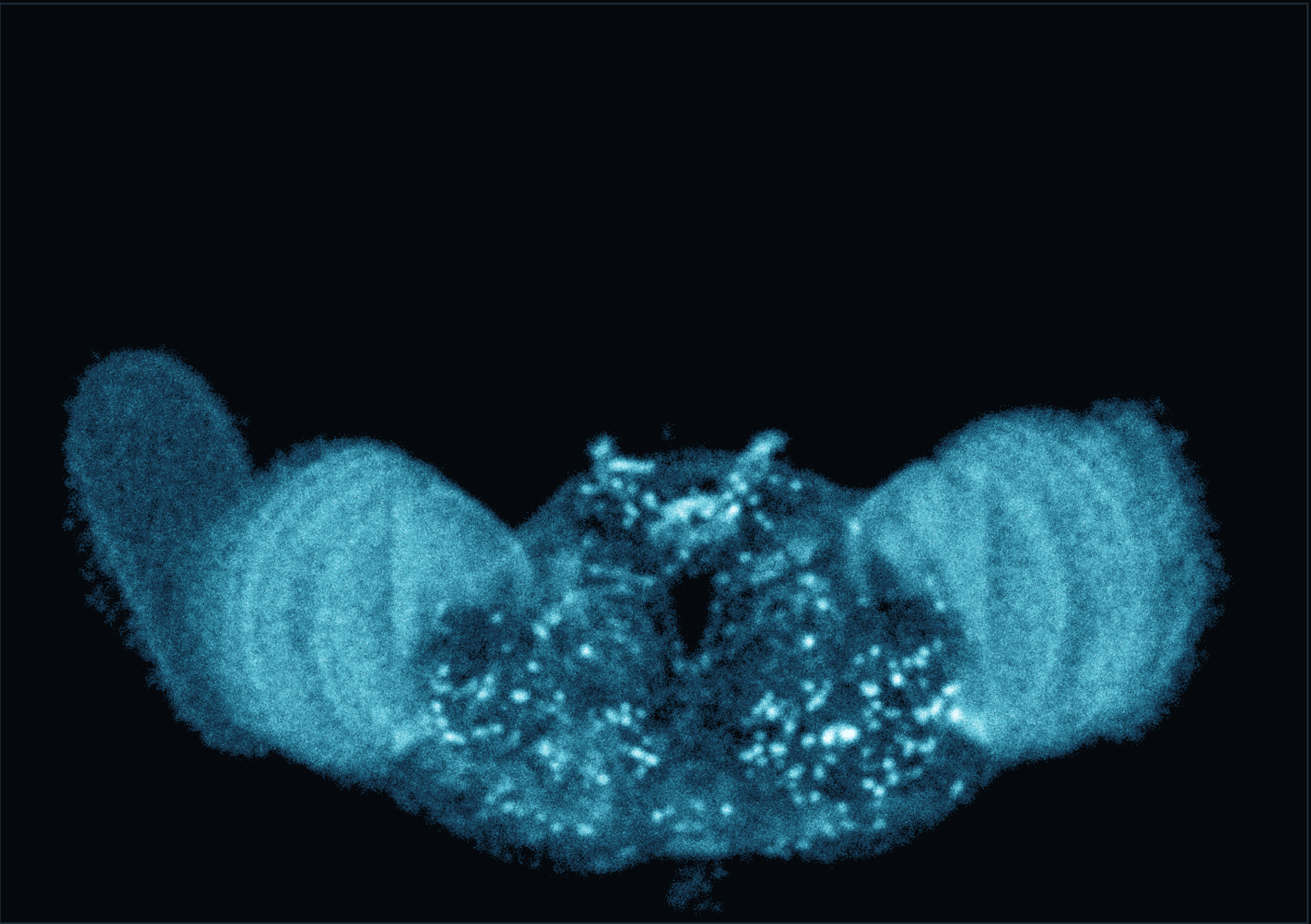


the real brain: FlyWire v783, 139,241 of 139,255 somata placed
0.29% of pixels hold a soma · log scale to its own ceiling 2 counts/px (busiest pixel 3)

10x
1,392,550 neurons · 27,842,102 connections



subdivided ladder top rung (10 children per soma)
2.74% of pixels hold a soma · log scale to its own ceiling 3 counts/px (busiest pixel 6)



Vyb/CUDA kernels; no pixel saturates, the panel resolves structure
16.30% of pixels hold a soma · log scale to its own ceiling 12 counts/px (busiest pixel 22)

1x, 10x and 100x are the device rasters from the Vyb NVPTX kernels in src/vyb_kernels/upscale_kernel.vyb (one atomic add per soma), not a re-plot; 0.1x is the phase-5 grouping, drawn as points. 2 x 5760 grid, 28.08 nm per pixel, each scaled to its own extent), but each density panel is tone-mapped to its own ceiling - printed on the panel - because occupancy differs by over an order of magnitude across the ladder (0.29% of pixels hold a soma at 1x, 2.74% at 10x, 16.30% at 100x): on one shared scale the 1x and 10x panels fall at 100x. Its busiest pixel holds 22 somata and the mean over occupied pixels is 1.81, so the panel resolves structure instead of reading as a silhouette; the ladder's real change is how much of the frame the cloud fills. Mean degree stays ~19 partners per neuron at every scale. Full resolution: [results/scale-gallery/zoom-100x.html](#) (or zoom in)